

Mohamed Ahmed

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Electrical Engineering | Hardware | Embedded Systems | Power Electronics | Controls | RF / Wireless

Education

University of Minnesota – Twin Cities

Expected May 2027

Bachelor of Science in Electrical Engineering

Relevant Coursework: Linear Control Systems & Laboratory, Power Electronics, Embedded Systems, Analog Electronics, Wireless Hardware System Design, Antenna Design

Technical Areas

Hardware & Analog Electronics: Mixed-Signal Hardware, Analog Front Ends, Transimpedance Amplifiers, Signal Conditioning, Sensor Interfaces, ADCs, Analog Filtering, Power Regulation, PCB Design, PCB Bring-Up

Embedded Systems: Embedded C/C++, FreeRTOS, ESP-IDF, ESP32-C3, STM32G4, Teensy 4.1, Interrupt-Driven Systems, ADC, PWM, Real-Time Sensor Acquisition

Digital & FPGA: Verilog, RTL, FPGA Development, Lattice ECP5, Digital Logic, Ethernet Hardware, Simulation, Synthesis

Power Electronics: DC-DC Conversion, Synchronous Buck Converters, MOSFET Power Stages, Gate Drivers, PWM, Voltage/Current Sensing, Power PCB Layout

Controls & Motor Drives: Feedback Control, PI/PID Control, DC Motor Modeling, Speed Control, Position Control, System Identification, Transfer Functions, Bode Analysis, Stability Margins

RF & Wireless: RF Receiver/Transmitter Architectures, Gain, Noise Figure, Linearity, Sensitivity, Dynamic Range, S-Parameters, Transmission Lines, Impedance Matching, Smith Charts, Antennas

Professional Experience

Michels Power

May 2026 – Aug 2026

Project Management Engineering Intern

- Supported electric utility projects through **field verification, GIS records, engineering drawings, and technical documentation**.
- Reviewed engineering drawings and as-built records for electric transmission and distribution projects.
- Developed and maintained a technical database tracking **1,200+ utility as-built records** for engineering and project documentation.
- Supported organization and verification of project records used across utility construction and engineering workflows.

Engineering Leadership

STRIVE Medical

September 2025 – Present

Electrical / Hardware Team Lead

- Lead electrical development for a **medical robotics platform** integrating sensing, actuation, embedded control, custom electronics, and power distribution.
- Designed a **4-layer PCBA** supporting embedded controllers, sensor interfaces, regulated power, and motor-drive hardware.
- Integrated **Teensy 4.1, ESP32, IMUs, 14-bit encoders, and motor controllers** across robotic subsystems.
- Developed embedded firmware for **sensor acquisition and motor-control interfaces**.
- Integrated embedded hardware using **I2C, SPI, UART, and BLE** interfaces.
- Performed PCB bring-up and hardware/software debugging using **oscilloscopes, logic analyzers, and multimeters**.
- Worked across hardware, firmware, sensing, and electromechanical interfaces during robotic system integration.
- Led electrical development contributing to **unanimous 1st Place** at the 2026 University of Minnesota MVP Challenge.

Engineering Projects

Wearable PPG Monitoring System

Mixed-Signal Hardware, Analog Front End, Embedded Systems, DSP

- Architected a mixed-signal wearable PCBA integrating a **24-bit ADC, optical analog front end, IMU, ESP32-C3, BLE**, and regulated power.
- Designed a **biased photodiode transimpedance amplifier** to convert low-level optical photocurrent into a measurable voltage signal.
- Designed analog signal-conditioning and filtering circuitry for low-frequency PPG acquisition prior to ADC conversion.

- Integrated analog, digital, sensing, and power subsystems on a custom PCB with attention to grounding, sensor interfaces, and low-noise signal routing.
- Developed **ESP-IDF** / **FreeRTOS firmware** for embedded acquisition and processing.
- Implemented **interrupt-driven SPI acquisition** using ADC data-ready events for deterministic sensor sampling.
- Synchronized PPG and IMU acquisition for real-time motion analysis and biosignal processing.
- Implemented a **101-tap FIR decimation pipeline** for digital signal processing.
- Implemented **IMU-referenced NLMS processing** for motion-artifact reduction.
- Developed **C and Python test software** to verify DSP precision, filter response, acquisition behavior, and embedded processing.
- Performed hardware/software bring-up and debugging using oscilloscopes, logic analyzers, C, and Python.

100 W Synchronous Buck Converter

Power Electronics, PCB Design, Modeling, Digital Feedback Control

- Designed a **100 W, 24 V-to-12 V synchronous buck converter** using N-channel MOSFETs, gate-drive circuitry, voltage/current sensing, and protection hardware.
- Developed averaged and **small-signal converter models** to characterize plant dynamics and closed-loop behavior.
- Derived the **control-to-output transfer function** and analyzed converter frequency response using MATLAB.
- Designed a **PI voltage-control loop** using Bode plots, crossover frequency, phase margin, and transient-response objectives.
- Performed extensive **LTspice simulation** of the switching power stage, sensing circuitry, transient behavior, and component operation.
- Used MATLAB and LTspice to compare analytical control models with switching-circuit behavior.
- Designed analog **voltage and current sensing interfaces** for an STM32G4-based digital controller.
- Developed an STM32G4 control architecture using **PWM generation, ADC feedback, and digital PI control**.
- Designed a **4-layer power PCB** with attention to high-current switching loops, gate-drive paths, grounding, sensing, and separation between power and control circuitry.

Feedback & Motor Control

MATLAB, Simulink, Quanser, Feedback Systems

- Modeled the dynamics of a **DC motor** and developed MATLAB/Simulink models for feedback-control analysis.
- Designed and tuned **PI speed controllers** for a DC motor.
- Evaluated closed-loop **stability, transient response, and robustness** using a Quanser DC Motor Control Trainer.
- Designed and tuned **PID position controllers** for motor position control.
- Investigated **integral windup** under actuator constraints.
- Applied encoder-based sensor feedback to electromechanical control problems.
- Performed **system identification** for a magnetic-levitation plant.
- Developed feedback-control approaches for a **nonlinear magnetic-levitation system**.

Four-Port Gigabit Ethernet Switch

FPGA, Verilog, Embedded Hardware

- Contributed to development of a **four-port Gigabit Ethernet switch** built around a Lattice ECP5 FPGA and four Gigabit Ethernet PHYs.
- Developed and integrated **Verilog RTL modules** supporting FPGA-based Ethernet functionality and hardware interfacing.
- Worked with **RGMII and MDIO interfaces** for FPGA-to-PHY communication, PHY configuration, and link management.
- Supported RTL verification and debugging through **simulation and synthesis**.
- Contributed to hardware-oriented test and debugging workflows.
- Contributed to a custom multi-layer PCB integrating the **FPGA, Ethernet PHYs, power rails, clocking, and external interfaces**.

Portable Neural Recording Device

Neural Sensing, Analog Front-End Research, Medical Hardware

- Collaborating with University of Minnesota clinicians and engineers on a portable system for recording **neural signals from implanted electrodes** outside the hospital.
- Investigating **low-noise amplification, signal conditioning, electrode interfaces, battery life, onboard storage, and wireless data transfer** for continuous recording.
- Evaluating hardware requirements for acquisition of small neural voltage signals while minimizing noise and motion-related interference.
- Supporting **early-stage system architecture and technology research** prior to prototype development.
- Contributing an electrical-engineering perspective to a multidisciplinary medical-device development effort involving clinical and engineering stakeholders.

3.5 GHz Helical Antenna

RF Design, ANSYS HFSS, Electromagnetics

- Designed and simulated an **axial-mode helical antenna centered near 3.5 GHz** using ANSYS HFSS for circularly polarized wireless operation.
- Performed parametric sweeps of **helix pitch and number of turns** to investigate antenna geometry and performance.
- Optimized antenna characteristics including **impedance matching, gain, bandwidth, and radiation behavior**.
- Analyzed **S11, VSWR, input impedance, radiation patterns, directivity, and realized gain**.
- Evaluated antenna behavior across approximately **2.5–4.0 GHz**.
- Achieved simulated peak gain of approximately **14 dBi** in optimized configurations.
- Compared return-loss and far-field simulation results across candidate antenna geometries.

Wireless Hardware System Design

RF Systems, Circuit Analysis, MATLAB

- Analyzed **RF receiver and transmitter architectures** and the function of LNAs, mixers, power amplifiers, and frequency-generation blocks within wireless transceivers.
- Evaluated RF system behavior using **gain, noise, linearity, bandwidth, sensitivity, and dynamic range**.
- Applied **S-parameters and transmission-line theory** to RF circuit analysis.
- Applied **impedance matching and Smith-chart techniques** to RF systems.
- Analyzed **noise figure and cascaded noise** in receiver signal chains.
- Analyzed RF nonlinearity including **harmonic distortion, intermodulation, and gain compression**.

Technical Skills

Programming: C, C++, Python, MATLAB, Verilog

Embedded Systems: Embedded C/C++, FreeRTOS, ESP-IDF, ESP32-C3, STM32G4, Teensy 4.1, Interrupts, ADC, PWM, GPIO

Digital Interfaces: SPI, I2C, UART, BLE

FPGA / Digital Design: Verilog, RTL, FPGA Development, Lattice ECP5, Digital Logic, Ethernet Hardware, Simulation, Synthesis

Analog Electronics: Analog Circuits, Transimpedance Amplifiers, Signal Conditioning, Sensor Interfaces, ADCs, Analog Filtering, Noise Analysis, Power Regulation

Power Electronics: DC-DC Converters, Synchronous Buck Converters, N-Channel MOSFETs, Gate Drivers, PWM, Voltage Sensing, Current Sensing, Switching Power Stages

Controls: Feedback Control, PI/PID Control, Transfer Functions, Small-Signal Modeling, Bode Analysis, Frequency-Domain Analysis, Stability Margins, System Identification

Motor Control & Mechatronics: DC Motor Modeling, Speed Control, Position Control, Encoders, Motor Controllers, Sensor Feedback, Electromechanical Systems

RF Systems: Receiver/Transmitter Architectures, Gain, Noise Figure, Linearity, Sensitivity, Dynamic Range

RF / Microwave: S-Parameters, Transmission Lines, Impedance Matching, Smith Chart, S11, VSWR, Radiation Patterns, Gain, Directivity

PCB Design: Schematic Capture, Altium Designer, KiCad, 4-Layer PCB Design, Mixed-Signal Layout, Power PCB Layout, Grounding, Sensor Interfaces, PCB Bring-Up

Test & Debug: Oscilloscope, Logic Analyzer, Multimeter, Hardware/Software Debugging, Root Cause Analysis, Hardware V&V

Modeling & Simulation: MATLAB, Simulink, LTspice, ANSYS HFSS, Dynamic Modeling, Small-Signal Modeling, Frequency-Domain Analysis

Engineering Tools: Altium Designer, KiCad, LTspice, MATLAB, Simulink, ANSYS HFSS, Git, SolidWorks

Selected Engineering Domains

Medical Devices: Biosignal Acquisition, Optical PPG Sensing, Neural Recording Hardware, Medical Robotics, Sensor Electronics, Wearable Electronics

Robotics & Mechatronics: Embedded Controllers, Motor Controllers, Encoders, IMUs, Sensor Integration, Feedback Control, Electromechanical Debugging

Power Conversion: Switching Converters, Power MOSFETs, Gate Drive, Digital Voltage Regulation, Power PCB Design, Sensing and Protection

Wireless & RF: RF System Analysis, Antennas, Impedance Matching, Transmission Lines, Receiver/Transmitter Architectures, Noise and Linearity Analysis

Embedded & Digital Hardware: Microcontrollers, FreeRTOS, Real-Time Sensor Acquisition, FPGA/Verilog, Digital Interfaces, Hardware/Firmware Integration

Engineering Recognition

2026 University of Minnesota MVP Challenge

Electrical development lead for STRIVE Medical's medical robotics platform; team received **unanimous 1st Place**.